No.	Type	Problem Statement
1	ML	**DATA PREPROCESSING** **Dataset:** Heart.csv Apply data preprocessing and exploratory data analysis on Heart.csv. Evaluate the dataset using summary statistics, and plot one visualization.
	AI	Solve the 8-Puzzle problem using the A* Search algorithm with the Manhattan Distance heuristic. Display the sequence of moves to reach the goal state.
2	ML	**REGRESSION: LINEAR REGRESSION AND CLASSIFICATION: LOGISTIC REGRESSION** **Dataset:** temperatures.csv, Social_Network_Ads.csv Apply data preprocessing. Train a Linear Regression model on temperatures.csv and a Logistic Regression model on Social_Network_Ads.csv. Evaluate both models using evaluation metrics, and plot one visualization.
	AI	Implement greedy Selection Sort to sort a numerical array in ascending order. Display the array state after each pass along with the final sorted array.
3	ML	**CLASSIFICATION: DECISION TREE CLASSIFIER** **Dataset:** Admission_Predict.csv Apply data preprocessing on Admission_Predict.csv. Train a Decision Tree Classifier to predict graduate admission. Evaluate the model using evaluation metrics, and plot one visualization.
	AI	Implement Dijkstra's algorithm to find the shortest path from a source vertex in a weighted graph. Display the shortest distances and paths.

No.	Type	Problem Statement
4	ML	**CLUSTERING: K-MEANS CLUSTERING AND HIERARCHICAL CLUSTERING** **Dataset:** Mall_Customers.csv Apply data preprocessing on Mall_Customers.csv. Implement both K-Means and Agglomerative Hierarchical Clustering on the dataset. Evaluate the models using evaluation metrics, and plot one visualization.
	AI	Implement Prim's algorithm using a min-heap to construct the Minimum Spanning Tree (MST) of a weighted graph. Display the selected edges and total minimum cost.
5	ML	**ASSOCIATION RULE MINING: APRIORI ALGORITHM** **Dataset:** Market_Basket_Optimization.csv Apply data preprocessing on Market_Basket_Optimization.csv. Implement the Apriori algorithm to discover frequent itemsets and association rules. Evaluate the rules using evaluation metrics, and plot one visualization.
	AI	Implement Kruskal's algorithm using the Disjoint-Set (Union-Find) data structure to construct the Minimum Spanning Tree (MST). Display the selected edges and total cost.
6	ML	**DATA PREPROCESSING** **Dataset:** melb_data.csv Apply data preprocessing and exploratory data analysis on melb_data.csv. Evaluate the dataset using summary statistics, and plot one visualization.
	AI	Implement the grid-based A* Search algorithm for obstacle avoidance and pathfinding. Display the optimal shortest path.
7	ML	**CLASSIFICATION: SUPPORT VECTOR MACHINE (SVM)** **Dataset:** UniversalBank.csv Apply data preprocessing on UniversalBank.csv. Train a Support Vector Machine (SVM) classifier to predict loan acceptance. Evaluate the model using evaluation metrics, and plot one visualization.
	AI	Implement Dijkstra's algorithm on a weighted network graph to find the minimum latency route from a source node to all destinations. Display the shortest paths and costs.

No.	Type	Problem Statement
8	ML	**CLUSTERING: K-MEANS CLUSTERING AND HIERARCHICAL CLUSTERING** **Dataset:** Cities_r2.csv Apply data preprocessing on Cities_r2.csv. Implement both K-Means and Agglomerative Hierarchical Clustering on the dataset. Evaluate the models using evaluation metrics, and plot one visualization.
	AI	Implement greedy Selection Sort to sort an array of numerical values. Display the minimum element selected and array state after each pass.
9	ML	**REGRESSION: LINEAR REGRESSION AND CLASSIFICATION: LOGISTIC REGRESSION** **Dataset:** Advertising.csv, UniversalBank.csv Apply data preprocessing. Train a Linear Regression model on Advertising.csv and a Logistic Regression model on UniversalBank.csv. Evaluate both models using evaluation metrics, and plot one visualization.
	AI	Implement Kruskal's algorithm using the Union-Find data structure with cycle detection to construct the MST for a weighted graph. Display the accepted edges and total cost.
10	ML	**CLASSIFICATION: DECISION TREE CLASSIFIER** **Dataset:** madfhantr.csv Apply data preprocessing on madfhantr.csv. Train a Decision Tree Classifier to predict loan eligibility. Evaluate the model using evaluation metrics, and plot one visualization.
	AI	Implement Prim's algorithm using a priority queue on a weighted graph. Display the sequence of selected edges and total minimum cost.
11	ML	**DATA PREPROCESSING** **Dataset:** sample_data.csv Apply data preprocessing and exploratory data analysis on sample_data.csv. Evaluate the dataset using summary statistics, and plot one visualization.
	AI	Implement the A* Search algorithm on a weighted graph using an admissible heuristic. Display the optimal path and total path cost.

No.	Type	Problem Statement
12	ML	**CLASSIFICATION: SUPPORT VECTOR MACHINE (SVM)** **Dataset:** Mobile Price dataset Apply data preprocessing on the Mobile Price dataset. Train a Support Vector Machine (SVM) classifier to predict price range. Evaluate the model using evaluation metrics, and plot one visualization.
	AI	Implement greedy Selection Sort to sort an array of benchmark scores in ascending order. Display intermediate array states and the final sorted array.
13	ML	**ASSOCIATION RULE MINING: APRIORI ALGORITHM** **Dataset:** Order1.csv Apply data preprocessing on Order1.csv. Implement the Apriori algorithm to discover frequent itemsets and association rules. Evaluate the rules using evaluation metrics, and plot one visualization.
	AI	Implement Prim's algorithm on a weighted graph using a priority queue to grow the tree from a source hub. Display the selected edges and final MST cost.
14	ML	**CLUSTERING: K-MEANS CLUSTERING AND HIERARCHICAL CLUSTERING** **Dataset:** 50_Startups.csv Apply data preprocessing on 50_Startups.csv. Implement both K-Means and Agglomerative Hierarchical Clustering on the dataset. Evaluate the models using evaluation metrics, and plot one visualization.
	AI	Implement Dijkstra's algorithm using a min-heap to determine the shortest path from a central source node to all destination nodes. Display the shortest distances and paths.
15	ML	**CLASSIFICATION: DECISION TREE CLASSIFIER** **Dataset:** pima-indians-diabetes.csv Apply data preprocessing on pima-indians-diabetes.csv. Train a Decision Tree Classifier to predict diabetes outcome. Evaluate the model using evaluation metrics, and plot one visualization.
	AI	Implement Kruskal's algorithm using Disjoint-Set (Union-Find) on a weighted network graph. Display candidate edge checks, selected MST edges, and total cost.

No.	Type	Problem Statement
16	ML	**DATA PREPROCESSING** **Dataset:** train.csv (Titanic dataset) Apply data preprocessing and exploratory data analysis on train.csv (Titanic dataset). Evaluate the dataset using summary statistics, and plot one visualization.
	AI	Implement the A* Search algorithm for grid pathfinding across obstacles from start to goal coordinates. Display the optimal path and traversal cost.
17	ML	**REGRESSION: LINEAR REGRESSION AND CLASSIFICATION: LOGISTIC REGRESSION** **Dataset:** Admission_Predict.csv, Social_Network_Ads.csv Apply data preprocessing. Train a Linear Regression model on Admission_Predict.csv and a Logistic Regression model on Social_Network_Ads.csv. Evaluate both models using evaluation metrics, and plot one visualization.
	AI	Implement Prim's algorithm using a min-heap on a weighted graph. Display the step-by-step selection of minimum-weight edges and total minimum cost.
18	ML	**ASSOCIATION RULE MINING: APRIORI ALGORITHM** **Dataset:** Order3.csv Apply data preprocessing on Order3.csv. Implement the Apriori algorithm to discover frequent purchase combinations. Evaluate the rules using evaluation metrics, and plot one visualization.
	AI	Implement greedy Selection Sort to sort an array of numerical values. Display the selected minimum element, intermediate array state after each pass, and final sorted list.
19	ML	**CLASSIFICATION: SUPPORT VECTOR MACHINE (SVM)** **Dataset:** Admission_Predict.csv Apply data preprocessing on Admission_Predict.csv. Train a Support Vector Machine (SVM) classifier to predict admission qualification. Evaluate the model using evaluation metrics, and plot one visualization.
	AI	Implement Kruskal's algorithm using Disjoint-Set (Union-Find) on a weighted graph. Display sorted edges, cycle checks, chosen MST edges, and total cost.

No.	Type	Problem Statement
20	ML	**ASSOCIATION RULE MINING: APRIORI ALGORITHM** **Dataset:** Retail Transaction dataset Apply data preprocessing on the Retail Transaction dataset. Implement the Apriori algorithm to discover association rules from transactions. Evaluate the rules using evaluation metrics, and plot one visualization.
	AI	Implement Dijkstra's algorithm using a priority queue on a weighted graph. Given a source hub, display shortest delivery distances and routes to all destination nodes.
21	ML	**DATA PREPROCESSING** **Dataset:** Movie-Rating.csv Apply data preprocessing and exploratory data analysis on Movie-Rating.csv. Evaluate the dataset using summary statistics, and plot one visualization.
	AI	Solve the 8-Puzzle problem using the A* Search algorithm with an admissible heuristic. Display the sequence of moves and total cost to reach the goal state.
22	ML	**REGRESSION: LINEAR REGRESSION AND CLASSIFICATION: LOGISTIC REGRESSION** **Dataset:** weight-height.csv, UniversalBank.csv Apply data preprocessing. Train a Linear Regression model on weight-height.csv and a Logistic Regression model on UniversalBank.csv. Evaluate both models using evaluation metrics, and plot one visualization.
	AI	Implement Dijkstra's algorithm using a priority queue on a weighted graph. Display minimum transit times and shortest paths from the source vertex to all vertices.
23	ML	**CLUSTERING: K-MEANS CLUSTERING AND HIERARCHICAL CLUSTERING** **Dataset:** hitters.csv Apply data preprocessing on hitters.csv. Implement both K-Means and Agglomerative Hierarchical Clustering on the dataset. Evaluate the models using evaluation metrics, and plot one visualization.
	AI	Implement Kruskal's algorithm using Disjoint-Set (Union-Find) on a weighted graph. Display sorted edges, cycle checks, accepted MST edges, and total cost.

No.	Type	Problem Statement
24	ML	**CLASSIFICATION: DECISION TREE CLASSIFIER** **Dataset:** Heart.csv Apply data preprocessing on Heart.csv. Train a Decision Tree Classifier to predict heart disease (AHD). Evaluate the model using evaluation metrics, and plot one visualization.
	AI	Implement greedy Selection Sort to sort an array of numerical scores. Display selected elements, intermediate states after each pass, and the final sorted array.
25	ML	**CLASSIFICATION: SUPPORT VECTOR MACHINE (SVM)** **Dataset:** Social_Network_Ads.csv Apply data preprocessing on Social_Network_Ads.csv. Train a Support Vector Machine (SVM) classifier to predict ad purchase (Purchased). Evaluate the model using evaluation metrics, and plot one visualization.
	AI	Implement Prim's algorithm using a min-heap on a weighted graph. Display each selected connection in sequence and the final minimum total cost.